



NIST AI 100-1



Artificial Intelligence Risk Management Framework (AI RMF 1.0)

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Update Schedule and Versions

The Artificial Intelligence Risk Management Framework (AI RMF) is intended to be a living document.

NIST will review the content and usefulness of the Framework regularly to determine if an update is appropriate; a review with formal input from the AI community is expected to take place no later than 2028. The Framework will employ a two-number versioning system to track and identify major and minor changes. The first number will represent the generation of the AI RMF and its companion documents (e.g., 1.0) and will change only with major revisions. Minor revisions will be tracked using “n” after the generation number (e.g., 1.1). All changes will be tracked using a Version Control Table which identifies the history, including version number, date of change, and description of change. NIST plans to update the AI RMF Playbook frequently. Comments on the AI RMF Playbook may be sent via email to AIframework@nist.gov at any time and will be reviewed and integrated on a semi-annual basis.

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Executive Summary

Artificial intelligence (AI) technologies have significant potential to transform society and people's lives – from commerce and health to transportation and cybersecurity to the environment and our planet. AI technologies can drive inclusive economic growth and support scientific advancements that improve the conditions of our world. AI technologies, however, also pose risks that can negatively impact individuals, groups, organizations, communities, society, the environment, and the planet. Like risks for other types of technology, AI risks can emerge in a variety of ways and can be characterized as long- or short-term, high- or low-probability, systemic or localized, and high- or low-impact.

The AI RMF refers to an *AI system* as an engineered or machine-based system that can, for a given set of objectives, generate outputs such as predictions, recommendations, or decisions influencing real or virtual environments. AI systems are designed to operate with varying levels of autonomy (Adapted from: OECD Recommendation on AI:2019; ISO/IEC 22989:2022).

While there are myriad standards and best practices to help organizations mitigate the risks of traditional software or information-based systems, the risks posed by AI systems are in many ways unique (See Appendix B). AI systems, for example, may be trained on data that can change over time, sometimes significantly and unexpectedly, affecting system functionality and trustworthiness in ways that are hard to understand. AI systems and the contexts in which they are deployed are frequently complex, making it difficult to detect and respond to failures when they occur. AI systems are inherently socio-technical in nature, meaning they are influenced by societal dynamics and human behavior. AI risks – and benefits – can emerge from the interplay of technical aspects combined with societal factors related to how a system is used, its interactions with other AI systems, who operates it, and the social context in which it is deployed.

These risks make AI a uniquely challenging technology to deploy and utilize both for organizations and within society. Without proper controls, AI systems can amplify, perpetuate, or exacerbate inequitable or undesirable outcomes for individuals and communities. With proper controls, AI systems can mitigate and manage inequitable outcomes.

AI risk management is a key component of responsible development and use of AI systems. Responsible AI practices can help align the decisions about AI system design, development, and uses with intended aim and values. Core concepts in responsible AI emphasize human centrality, social responsibility, and sustainability. AI risk management can drive responsible uses and practices by prompting organizations and their internal teams who design, develop, and deploy AI to think more critically about context and potential or unexpected negative and positive impacts. Understanding and managing the risks of AI systems will help to enhance trustworthiness, and in turn, cultivate public trust.

Social responsibility can refer to the organization’s responsibility “for the impacts of its decisions and activities on society and the environment through transparent and ethical behavior” (ISO 26000:2010). *Sustainability* refers to the “state of the global system, including environmental, social, and economic aspects, in which the needs of the present are met without compromising the ability of future generations to meet their own needs” (ISO/IEC TR 24368:2022). Responsible AI is meant to result in technology that is also equitable and accountable. The expectation is that organizational practices are carried out in accord with “*professional responsibility*,” defined by ISO as an approach that “aims to ensure that professionals who design, develop, or deploy AI systems and applications or AI-based products or systems, recognize their unique position to exert influence on people, society, and the future of AI” (ISO/IEC TR 24368:2022).

As directed by the National Artificial Intelligence Initiative Act of 2020 (P.L. 116-283), the goal of the AI RMF is to offer a resource to the organizations designing, developing, deploying, or using AI systems to help manage the many risks of AI and promote trustworthy and responsible development and use of AI systems. The Framework is intended to be **voluntary**, rights-preserving, non-sector-specific, and use-case agnostic, providing flexibility to organizations of all sizes and in all sectors and throughout society to implement the approaches in the Framework.

The Framework is designed to equip organizations and individuals – referred to here as *AI actors* – with approaches that increase the trustworthiness of AI systems, and to help foster the responsible design, development, deployment, and use of AI systems over time. AI actors are defined by the Organisation for Economic Co-operation and Development (OECD) as “those who play an active role in the AI system lifecycle, including organizations and individuals that deploy or operate AI” [OECD (2019) Artificial Intelligence in Society—OECD iLibrary] (See Appendix A).

The AI RMF is intended to be practical, to adapt to the AI landscape as AI technologies continue to develop, and to be operationalized by organizations in varying degrees and capacities so society can benefit from AI while also being protected from its potential harms.

The Framework and supporting resources will be updated, expanded, and improved based on evolving technology, the standards landscape around the world, and AI community experience and feedback. NIST will continue to align the AI RMF and related guidance with applicable international standards, guidelines, and practices. As the AI RMF is put into use, additional lessons will be learned to inform future updates and additional resources.

The Framework is divided into two parts. Part 1 discusses how organizations can frame the risks related to AI and describes the intended audience. Next, AI risks and trustworthiness are analyzed, outlining the characteristics of trustworthy AI systems, which include

valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy enhanced, and fair with their harmful biases managed.

Part 2 comprises the “Core” of the Framework. It describes four specific functions to help organizations address the risks of AI systems in practice. These functions – **GOVERN**, **MAP**, **MEASURE**, and **MANAGE** – are broken down further into categories and subcategories. While **GOVERN** applies to all stages of organizations’ AI risk management processes and procedures, the **MAP**, **MEASURE**, and **MANAGE** functions can be applied in AI system-specific contexts and at specific stages of the AI lifecycle.

Additional resources related to the Framework are included in the AI RMF Playbook, which is available via the NIST AI RMF website:

<https://www.nist.gov/itl/ai-risk-management-framework>.

Development of the AI RMF by NIST in collaboration with the private and public sectors is directed and consistent with its broader AI efforts called for by [the National AI Initiative Act of 2020](#), [the National Security Commission on Artificial Intelligence recommendations](#), and [the Plan for Federal Engagement in Developing Technical Standards and Related Tools](#). Engagement with the AI community during this Framework’s development – via responses to a formal Request for Information, three widely attended workshops, public comments on a concept paper and two drafts of the Framework, discussions at multiple public forums, and many small group meetings – has informed development of the AI RMF 1.0 as well as AI research and development and evaluation conducted by NIST and others. Priority research and additional guidance that will enhance this Framework will be captured in an associated AI Risk Management Framework Roadmap to which NIST and the broader community can contribute.

Part 1: Foundational Information

1. Framing Risk

AI risk management offers a path to minimize potential negative impacts of AI systems, such as threats to civil liberties and rights, while also providing opportunities to maximize positive impacts. Addressing, documenting, and managing AI risks and potential negative impacts effectively can lead to more trustworthy AI systems.

1.1 Understanding and Addressing Risks, Impacts, and Harms

In the context of the AI RMF, *risk* refers to the composite measure of an event’s probability of occurring and the magnitude or degree of the consequences of the corresponding event. The impacts, or consequences, of AI systems can be positive, negative, or both and can result in opportunities or threats (Adapted from: ISO 31000:2018). When considering the negative impact of a potential event, risk is a function of 1) the negative impact, or magnitude of harm, that would arise if the circumstance or event occurs and 2) the likelihood of occurrence (Adapted from: OMB Circular A-130:2016). Negative impact or harm can be experienced by individuals, groups, communities, organizations, society, the environment, and the planet.

“Risk management refers to coordinated activities to direct and control an organization with regard to risk” (Source: ISO 31000:2018).

While risk management processes generally address negative impacts, this Framework offers approaches to minimize anticipated negative impacts of AI systems *and* identify opportunities to maximize positive impacts. Effectively managing the risk of potential harms could lead to more trustworthy AI systems and unleash potential benefits to people (individuals, communities, and society), organizations, and systems/ecosystems. Risk management can enable AI developers and users to understand impacts and account for the inherent limitations and uncertainties in their models and systems, which in turn can improve overall system performance and trustworthiness and the likelihood that AI technologies will be used in ways that are beneficial.

The AI RMF is designed to address new risks as they emerge. This flexibility is particularly important where impacts are not easily foreseeable and applications are evolving. While some AI risks and benefits are well-known, it can be challenging to assess negative impacts and the degree of harms. Figure 1 provides examples of potential harms that can be related to AI systems.

AI risk management efforts should consider that humans may assume that AI systems work – and work well – in *all* settings. For example, whether correct or not, AI systems are often perceived as being more objective than humans or as offering greater capabilities than general software.



Fig. 1. Examples of potential harms related to AI systems. Trustworthy AI systems and their responsible use can mitigate negative risks and contribute to benefits for people, organizations, and ecosystems.

1.2 Challenges for AI Risk Management

Several challenges are described below. They should be taken into account when managing risks in pursuit of AI trustworthiness.

1.2.1 Risk Measurement

AI risks or failures that are not well-defined or adequately understood are difficult to measure quantitatively or qualitatively. The inability to appropriately measure AI risks does not imply that an AI system necessarily poses either a high or low risk. Some risk measurement challenges include:

Risks related to third-party software, hardware, and data: Third-party data or systems can accelerate research and development and facilitate technology transition. They also may complicate risk measurement. Risk can emerge both from third-party data, software or hardware itself and how it is used. Risk metrics or methodologies used by the organization developing the AI system may not align with the risk metrics or methodologies used by the organization *deploying or operating* the system. Also, the organization developing the AI system may not be transparent about the risk metrics or methodologies it used. Risk measurement and management can be complicated by how customers use or integrate third-party data or systems into AI products or services, particularly without sufficient internal governance structures and technical safeguards. Regardless, all parties and AI actors should manage risk in the AI systems they develop, deploy, or use as standalone or integrated components.

Tracking emergent risks: Organizations' risk management efforts will be enhanced by identifying and tracking emergent risks and considering techniques for measuring them.